Dmitry B. Eremin

University of Texas at Austin
Department of Chemistry
105 East 24th Street, Austin, TX 78712, U.S.
e-mail: eremin@utexas.edu

Professional Positions

01/2027	***Assistant Professor***, University of Texas at Austin, Department of Chemistry
2024-present	***Staff Scientist***, California Institute of Technology, Division of Chemistry and Chemical Engineering, Prof. Hosea Nelson group
2024-present	***Visiting Scientist***, University of California, Los Angeles, Department of Chemistry and Biochemistry, Prof. Jose Rodriguez group
2019-2024	***Agilent Postdoctoral Fellow***, University of Southern California Mentor: Prof. Valery Fokin

Education

2014-2019 *(Ph.D.)*	Zelinsky Institute of Organic Chemistry, Supervisor: Prof. Valentine Ananikov Thesis title: *"The Key Role of Metal-Ligand Bond Lability in the Formation of Active Centers in Palladium-Catalyzed Reactions"*
2009-2014 *(M.Sc.)*	Lomonosov Moscow State University, Faculty of Chemistry, *Summa cum laude* M.Sc. thesis title: *"Studying Structure and Reactivity of Transition Metal Complexes Using Electrospray Ionization High Resolution Mass Spectrometry"* Supervisor: Prof. Valentine Ananikov

Awards and Honors

- CAS Future Leaders, American Chemical Society, 2025
- Top 100 CAS Future Leaders, American Chemical Society, 2024
- Top 100 CAS Future Leaders, American Chemical Society, 2023
- The Mistletoe Research Fellowship, 2023-2024 Waitlist Candidate, Momental Foundation, 2023
- Arthur W. Adamson Postdoctoral Recognition Award, University of Southern California, 2023
- RSC Researcher Development and Travel Grant, Royal Society of Chemistry, 2023
- Anton B. Burg Foundation Postdoctoral Teaching Award, University of Southern California, 2022
- Travel Grant for PhD Students and Early Career Scientists, Royal Society of Chemistry, 2019
- Agilent Postdoctoral Fellowship, Agilent Technologies, 2019
- Gold medal of Russian Academy of Sciences for young researchers (field of chemistry), 2018
- Laureate, Young researcher competition, 2018 Winter School in Organic Chemistry, Moscow region, 2018
- 2^{nd} Prize, Young researcher competition, 8^{th} RSMS meeting, Moscow, 2017
- A.N. Nesmeyanov scholarship for PhD students, Zelinsky Institute of Organic Chemistry, 2017-2018
- Laureate, Tutor competition, E-NANOXI Olympiad, Lomonosov Moscow State University, 2017
- 1^{st} Prize, Young researcher competition, 7^{th} RSMS meeting, Moscow, 2015
- A.N. Nesmeyanov scholarship for PhD students, Zelinsky Institute of Organic Chemistry, 2015-2016
- The best course project in Analytical Chemistry, Lomonosov Moscow State University, 2011
- Crystal Owl recipient, "What? Where? When?" series

Publications

Full list of publications available via Web of Science ResearcherID: [R-3730-2016](R-3730-2016) or ORCID: [0000-0003-2946-5293](0000-0003-2946-5293)

(1) Parsons, S. W.; Judd, K. D.; Eremin, D. B.; Fokin, V. V.; Dawlaty, J. M. "Mapping Nonlinear Solvation at the Air-Organic-Solution Interface Using an Azide Probe" *J. Phys. Chem. Lett.* **2026**, *17*, 1192-1200.

(2) Parsons, S. W.; Judd, K. D.; **Eremin, D. B.**; Fokin, V. V.; Dawlaty, J. M. "Where's the Salt? Sensing the Ionic Environment of the Air–Organic–Water Interface Using an Azide Probe" *Angew. Chem. Int. Ed.* **2026**, *65*, e202518462.

(3) **Eremin, D. B.**; Jha, K. K.; Delgadillo, D. A.; Zhang, H.; Foxman, S. H.; Johnson, S. N.; Vlahakis, N. W.; Cascio, D.; Lavallo, V.; Rodríguez, J. A.; Nelson, H. M. "Spatially-Aware Diffraction Mapping Enables Fully Autonomous MicroED" *J. Am. Chem. Soc.* **2025**, *147*, 42299-42310.

(4) Judd, K. D.; Parsons, S. W.; **Eremin, D. B.**; Fokin, V. V.; Dawlaty, J. M. "Hydrolysis Kinetics at the Air–Water Interface" *J. Am. Chem. Soc.* **2025**, *147*, 31741–31748.

(5) Majumder, T.; **Eremin, D. B.**; Delibas, B.; Sarkar, A.; Fokin, V. V.; Dawlaty, J. M. "Calibrating the Oxidative Capacity of Microdroplets" *Angew. Chem. Int. Ed.* **2025**, *64,* e202414746.

(6) Judd, K. D.; Parsons, S. W.; **Eremin, D. B.**; Fokin, V. V.; Dawlaty, J. M. "Visualizing partial solvation at the air–water interface" *Chem. Sci.* **2024**, *15*, 8346–8354.

(7) Grotsch, K.; Sadybekov, A. V.; Hiller, S. R.; Zaidi, S.; **Eremin, D. B.**; Le, A.; Liu, Y.; Smith, E. C.; Illiopoulis-Tsoutsouvas, C.; Thomas, J.; Aggarwal, S.; Pickett, J. E.; Reyes, C.; Picazo, E.; Roth, B. L.; Makryannis, A.; Katritch, V.; Fokin, V. V. "Virtual Screening of a Chemically Diverse "Superscaffold" Library Enables Ligand Discovery for a key GPCR Target" *ACS Chem. Bio.* **2024**, *19,* 866–874.

(8) Patil, E. D.; Burykina, J. V.; **Eremin, D. B.**; Boiko, D. A.; Shepelenko, K. E.; Ilyushenkova, V. V.; Chernyshev, V. M.; Ananikov, V. P. "Quantitative Determination of Active Species Transforming the R-NHC Coupling Process under Catalytic Conditions" *Inorg. Chem.* **2024**, *63*, 2967–2976.

(9) Aggarwal, S.; Vu, A.; **Eremin, D. B.**; Persaud, R.; Fokin, V. V. "Arenes Participate in 1,3-Dipolar Cycloaddition with *in situ* Generated Diazoalkenes" *Nat. Chem.* **2023**, *15*, 764-772. (Featured: Front cover, *Synfacts*)

(10) Galushko, A. S.; Boiko, D. A.; Pentsak, E. O.; **Eremin, D. B.**; Ananikov, V. P. "Time-Resolved Formation and Operation Maps of Pd Catalysts Suggest a Key Role of Single Atom Centers in Cross-Coupling" *J. Am. Chem. Soc.* **2023**, *145*, 9092-9103.

(11) Sahharova, L. T.; Burykina, J. V.; Kostyukovich, A. Yu.; **Eremin, D. B.**; Boiko, D. A.; Fakhrutdinov, A. N.; Ananikov, V. P. "Expanding the Role of Dimeric Species: On-Cycle Involvement, Improved Stability, and Control of Stereo-Specificity. A Case Study of Atom-Economic Catalytic Hydrothiolation" *ACS Catal.* **2023**, *13*, 3591-3604.

(12) Strumolo, M. J.; **Eremin, D. B.**; Wang, S.; Perez, C. M.; Prezhdo, O. V.; Figueroa, J. S.; Brutchey, R. L. "Formation of a P_{16}^{2-} Ink from Elemental Red Phosphorus in a Thiol–Amine Mixture" *Inorg. Chem.* **2023**, *62*, 6197-6201.

(13) **Eremin, D. B.**;[‡] Galushko, A. S.;[‡] Boiko, D. A.;[‡] Pentsak, E. O.; Chistyakov, I. V.; Ananikov, V. P. "Towards Totally Defined Nanocatalysis: Deep Learning Reveals the Extraordinary Activity of Single Pd/C Particles" *J. Am. Chem. Soc.* **2022**, *144*, 6071-6079. ([‡]authors contributed equally; Featured: JACS Spotlights)

(14) **Eremin, D. B.**; Fokin, V. V. "Dual Electrospray Ionization Enhancement of Proteins Enabled by DMSO Supercharging Reagent" *J. Am. Soc. Mass Spectrom.* **2022**, *33*, 203-206.

(15) Chenyshev, V. M.; Khazipov, O. V.; Shevchenko, M. A.; Pasyukov, D. V.; Burykina, J. V.; Minyaev, M. E.; **Eremin, D. B.**; Ananikov, V. P. "Discovery of the N−NHC Coupling Process under the Conditions of Pd/NHC- and Ni/NHC-Catalyzed Buchwald−Hartwig Amination" *Organometallics* **2022**, *41*, 1519-1531.

(16) Koskela, K. M.; Perez, C. M.; **Eremin, D. B.**; Evans, J. M.; Strumolo, M. J.; Lewis, N. S.; Prezhdo, O. V.; Brutchey, R. L. "Polymorphic Control of Solution-Processed Cu_2SnS_3 Films with Thiol−Amine Ink Formulation" *Chem. Mater.* **2022**, *34*, 8654-8663.

(17) Eremina, O. E.; Czaja, A.; Fernando, A.; Aron, A.; **Eremin, D. B.**; Zavaleta, C. "Expanding the Multiplexing Capabilities of Raman Imaging to Reveal Highly Specific Molecular Expression and Enable Spatial Profiling" *ACS Nano* **2022**, *16*, 10341-10353.

(18) Eremina, O. E.; Kapitanova, O. O.; Ferree, M. V.; Lemesh, I. A.; **Eremin, D. B.**; Goodilin, E. A.; Veselova, I. A. "Ultrasensitive and Multiplex SERS Determination of Anthropogenic Phenols in Oil Fuel and Environmental Samples" *Environ. Sci.: Nano,* **2022**, *9*, 964-974.

(19) **Eremin, D. B.**; Fokin, V. V. "On-Water Selectivity Switch in Microdroplets in the 1,2,3-Triazole Synthesis from Bromoethenesulfonyl Fluoride" *J. Am. Chem. Soc.* **2021**, *143*, 18374-18379. (Featured: JACS Spotlights, Front cover)

(20) Miyamoto, Y.; Aggarwal, S.; Celaje, J. J. A.; Ihara, S.; Ang, J.; **Eremin, D. B.**; Land, K. M.; Wrischnik, L. A.; Zhang, L.; Fokin, V. V.; Eckmann, L. "Gold(I) Phosphine Derivatives with Improved Selectivity as Topically Active Drug Leads to Overcome 5-Nitroheterocyclic Drug Resistance in Trichomonas vaginalis" *J. Med. Chem.* **2021**, *64*, 6608-6620.

(21) Kostyukovich, A. Yu.; Burykina, J. V.; **Eremin, D. B.;** Ananikov, V. P. "Detection and Structural Investigation of Elusive Palladium Hydride Intermediates Formed from Simple Metal Salts" *Inorg. Chem.* **2021**, *60*, 7128-7142.

(22) Chernyshev, V. M.; Khazipov, O. V.; Denisova, E. A.; **Eremin, D. B.**; Ananikov, V. P. "Formation and Stabilization of Nanosized Pd Particles in Catalytic Systems: Ionic Nitrogen Compounds as Catalytic Promoters and Stabilizers of Nanoparticles" *Coord. Chem. Rev.* **2021**, *437,* 213860.

(23) Eremina, O. E.; **Eremin, D. B.**; Czaja, A.; Zavaleta, C. "Selecting Surface-Enhanced Raman Spectroscopy Flavors for Multiplexed Imaging Applications: Beyond the Experiment" *J. Phys. Chem. Lett.* **2021**, *12*, 5564-5570.

(24) Runikhina, S. A.; **Eremin, D. B.**; Chusov, D. "Reductive Aldol-Type Reactions in the Synthesis of Pharmaceuticals" *Chem. Eur. J.* **2021**, *27*, 15327-15360. (Featured: Frontispiece)

(25) Sahharova, L. T.; Gordeev, E. G.; **Eremin, D. B.**; Ananikov, V. P. "Pd-Catalyzed Synthesis of Densely Functionalized Cyclopropyl Vinyl Sulfides Reveals the Origin of High Selectivity in a Fundamental Alkyne Insertion Step" *ACS Catal.* **2020**, *10*, 9872-9888.

(26) **Eremin, D. B.;** Boiko, D. A.; Kostyukovich, A. Yu.; Burykina, J. V.; Denisova, E. A.; Anania, M.; Martens, J.; Berden, G.; Oomens, J.; Roithová, J.; Ananikov, V. P. "Mechanistic Study of Pd/NHC-Catalyzed Sonogashira Reaction: Discovery of NHC-Ethynyl Coupling Process" *Chem. Eur. J.*, **2020**, *26*, 15672-15681.

(27) Lowe, J. C.; Wright, L. D.; **Eremin, D. B.**; Burykina, J. V.; Martens, J.; Plasser, F.; Ananikov, V. P.; Bowers, J. W.; Malkov, A. V. "Solution Processed CZTS Solar Cells Using Amine-Thiol Systems: Understanding the Dissolution Process and Device Fabrication" *J. Mater. Chem. C* **2020**, *8*, 10309-10318.

(28) Chernyshev, V. M.; Denisova, E. A.; **Eremin, D. B.**; Ananikov, V. P. "The Key Role of R-NHC Couplings (R = C, H, Heteroatom) and M-NHC Bond Cleavage in the Evolution of M/NHC Complexes and Formation of Catalytically Active Species" *Chem. Sci.* **2020**, *11* 6957-6977.

(29) Burykina, J. V.; Boiko, D. A.; Ilyushenkova, V. V.; **Eremin, D. B.**; Ananikov, V. P. "Comprehensive Mass Spectrometric Mapping of Chemical Compounds for the Development of Algorithms for Machine Learning and Artificial Intelligence" *Doklady Phys. Chem.*, **2020**, *492*, 51-56.

(30) **Eremin, D. B.**; Denisova, E. A.; Kostyukovich, A. Yu.; Martens, J.; Berden, G.; Oomens, J.; Khrustalev, V. N.; Chernyshev, V. M.; Ananikov, V. P. "Ionic Pd/NHC Catalytic System Enables Recoverable Homogeneous Catalysis. Mechanistic Study and Application in the Mizoroki-Heck Reaction" *Chem. Eur. J.* **2019**, *72*, 16564-16572. (Featured: Front cover, Cover profile)

(31) Pentsak, E. O.; **Eremin, D. B.**; Gordeev, E. G.; Ananikov, V. P. "Phantom Reactivity in Organic and Catalytic Reactions as a Consequence of Microscale Destruction and Contamination-Trapping Effects of Magnetic Stir Bars" *ACS Catal.* **2019**, *9*, 3070-3081.

(32) Chernyshev, V. M.; Astakhov, A. V.; Chikunov, I. E.; Tyurin, R. V.; **Eremin, D. B.**; Ranny, G. S.; Khrustalev, V. N.; Ananikov, V. P. "Pd and Pt Catalyst Poisoning in the Study of Reaction Mechanisms: What Does the Mercury Test Mean for Catalysis?" *ACS Catal.* **2019**, *9*, 2984-2995.

(33) Titov, K.;‡ **Eremin, D. B.**;‡ Kashin, A. S.; Boada, R.; Souza, B. E.; Keley, C. S.; Frogley, M. D.; Cinque, G; Gianolio, D.; Cibin, G.; Rudić, S.; Ananikov, V. P.; Tan, J. C. "OX-1 Metal-Organic Framework Nanosheets as Robust Hosts for Highly Active Catalytic Palladium Species" *ACS Sustainable Chem. Eng.* **2019**, *7*, 5875-5885. (‡authors contributed equally)

(34) Denisova, E. A.; **Eremin, D. B.**; Gordeev, E. G.; Tsedilin, A. M.; Ananikov, V. P. “Addressing Reversibility of R-NHC Coupling on Palladium: Is Nano-to-Molecular Transition Possible for Pd/NHC System?” *Inorg. Chem.* **2019**, *58*, 12218-12227.

(35) Kostyukovich, A. Yu.; Tsedilin, A. M.; Sushchenko, E. D.; **Eremin, D. B.**; Kashin, A. S.; Topchiy, M. A.; Asachenko, A. F.; Nechaev, M. S.; Ananikov, V. P. “*In situ* Transformations of Pd/NHC Complexes to Colloidal Pd Nano-Particles Studied for N-Heterocyclic Carbene Ligands of Different Nature” *Inorg. Chem. Front.* **2019**, *6*, 482-492.

(36) Egorova, K. S.; Sinjushin, A. A.; Posvyatenko, A. V.; **Eremin, D. B.**; Kashin, A. S.; Galushko, A. S.; Ananikov, V. P. “Evaluation of Phytotoxicity and Cytotoxicity of Industrial Catalyst Components (Fe, Cu, Ni, Rh and Pd): A Case of Lethal Toxicity of a Rhodium Salt in Terrestrial Plants” *Chemosphere* **2019**, *223*, 738-747.

(37) Larionov, V. A.; Yashkina, L. V.; Medvedev, M. G.; Smol’yakov, A. F.; Peregudov, A. S.; Pavlov, A. A.; **Eremin, D. B.**; Savel’yeva, T. F.; Maleev, V. I.; Belokon, Yu. N. “Henry Reaction Revisited. Crucial Role of Water in an Asymmetric Henry Reaction Catalyzed by Chiral NNO-Type Copper(II) Complexes” *Inorg. Chem.* **2019**, *58*, 11051-11065.

(38) Kashin, A. S.; Degtyareva, E. S.; **Eremin, D. B.**; Ananikov, V. P. “Exploring the Performance of Nanostructured Reagents with Organic-group-defined Morphology in Cross-Coupling Reaction” *Nat. Commun.*, **2018**, *9*, 2396.

(39) Chenyshev, V. M.; Khazipov, O. V.; Shevchenko, M. A.; Chernenko, A. Yu.; Astakhov, A. V.; **Eremin, D. B.**; Pasyukov, D. V.; Kashin, A. S.; Ananikov, V. P. “Revealing the Unusual Role of Bases in Activation/Deactivation of Catalytic Systems: O–NHC Coupling in M/NHC Catalysis” *Chem. Sci.*, **2018**, *9*, 5564-5577.

(40) **Eremin, D. B.**; Boiko, D. A.; Borkovskaya, E. V.; Khrustalev, V. N.; Chenyshev, V. M.; Ananikov, V. P. “Ten-fold Boost of Catalytic Performance in Thiol-yne Click Reaction Enabled by Palladium Diketonate Complex with Hexafluoroacetylacetonate Ligand” *Catal. Sci. Tech.*, **2018**, *8*, 3073-3080.

(41) Khazipov, O. V.; Shevchenko, M. A.; Chernenko, A. Yu.; Astakhov, A. V.; Pasyukov, D. V.; **Eremin, D. B.**; Zubavichus, Y. V.; Khrustalev, V. N.; Chenyshev, V. M.; Ananikov, V. P. “Fast and slow release of catalytically active species in Metal/NHC systems induced by aliphatic amines” *Organometallics* **2018**, *37*, 1483-1492.

(42) Gordeev, E. G.; **Eremin, D. B.**; Chernyshev, V. M.; Ananikov, V. P. “Influence of R–NHC Coupling on the Outcome of R–X Oxidative Addition to Pd/NHC Complexes (R = Me, Ph, Vinyl, Ethynyl)” *Organometallics* **2018**, *37*, 787-796.

(43) Abakumov, G. A.; Piskunov, A. V.; Cherkasov, V. K.; Fedushkin, I. L.; Ananikov, V. P.; **Eremin, D. B.**; Gordeev, E. G.; Beletskaya, I. P.; Averin, A. D.; Bochkarev, M. N.; Trofonov, A. A.; Dgemilev, U. M.; Djakonov, V. A.; Egorov, M. P.; Vereschagin, A. N.; Syroeshkin, M. A.; Juykov, V. V.; Muzafarov, A. M.; Anisimov, A. A.; Arzumanyan, A. V.; Kononevich, Yu. N.; Temnikov, M. N.; Sinyashin, O. G.; Budnikova, Yu. G.; Burilov, A. R.; Karasik, A. A.; Mironov, V. F.; Storogenko, P. A.; Scherbakova, V. I.; Trofimov, B. A.; Amosova, S. V.; Gusarova, N. K.; Potapov, V. A.; Shur, V. B.; Burlakov, V. V.; Bogdanov, V. S.; Andreev, M. V. “Organoelement chemistry: promising areas of growth and challenges” *Russ. Chem. Rev.* **2018**, *87*, 393-507.

(44) Levitskiy, O. A.; Dulov, D. A.; Nikitin, O. M.; Bogdanov, A. V.; **Eremin, D. B.**; Paseshnichenko, K. A.; Magdesieva, T. V. “Competitive Routes for Electrochemical Oxidation of Substituted Diarylamines: the Guidelines” *ChemElectroChem* **2018**, *5*, 3391-3410.

(45) **Eremin, D. B.**; Ananikov, V. P. “Understanding active species in catalytic transformations: From molecular catalysis to nanoparticles, leaching, "Cocktails" of catalysts and dynamic systems” *Coord. Chem. Rev.* **2017**, *346*, 2-19. (Web of Science: Hot Paper 01/2018-07/2018; Highly Cited Paper since 01/2018)

(46) Kucherov, F. A.; Egorova, K. S.; Posvyatenko, A. V.; **Eremin, D. B.**; Ananikov, V. P. “Investigation of Cytotoxic Activity of Mitoxantrone at the Individual Cell Level by Using Ionic-Liquid-Tag-Enhanced Mass Spectrometry” *Anal. Chem.* **2017**, *89*, 13374-13381.

(47) Levitskiy, O. A.; **Eremin, D. B.**; Bogdanov, A. V.; Magdesieva, T. V. “Twisted Diarylnitroxides: An Efficient Route for Radical Stabilization” *Eur. J. Org. Chem.* **2017**, *2017*, 4726-4735.

(48) Ananikov, V. P.; **Eremin, D. B.**; Yakukhnov, S. A.; Dilman, A. D.; Levin, V. V.; Egorov, M. P.; Karlov, S. S.; Kustov, L. M.; Tarasov, A. L.; Greish, A. A.; Shesterkina, A. A.; Sakharov, A. M.; Nysenko, Z. N.; Sheremetev, A. B.; Stakheev, A. Y.; Mashkovsky, I. S.; Sukhorukov, A. Y.; Ioffe, S. L.; Terent'ev, A. O.; Vil, V. A.; Tomilov, Y. V.;

Novikov, R. A.; Zlotin, S. G.; Kucherenko, A. S.; Ustyuzhanina, N. E.; Krylov, V. B.; Tsvetkov, Y. E.; Gening, M. L.; Nifantiev, N. E. "Organic and hybrid systems: from science to practice" *Mendeleev Commun.* **2017**, *27*, 425-438.
(49) **Eremin, D. B.**; Kadentsev, V. I.; Sandulenko, I. V.; Moiseev, S. K. "Tandem high-resolution electrospray ionization mass spectrometry of fluorinated thevinols and 18,19-dihydrothevinols" *J. Anal. Chem.* **2015**, *70*, 1561-1568.
(50) Tsedilin, A. M.; Fakhrutdinov, A. N.; **Eremin, D. B.**; Zalesskiy, S. S.; Chizhov, A. O.; Kolotyrkina, N. G.; Ananikov, V. P. "How sensitive and accurate are routine NMR and MS measurements?" *Mendeleev Commun.* **2015**, *25*, 454-456.
(51) **Eremin, D. B.**; Ananikov, V. P. "Exceptional Behavior of Ni_2O_2 Species Revealed by ESI-MS and MS/MS Studies in Solution. Application of Superatomic Core to Facilitate New Chemical Transformations" *Organometallics* **2014**, *33*, 6353-6357.
(52) **Eremin, D. B.**; Banaru, A. M. "Multivariate Analysis of Melting Points for Homologous Series $C_xH_{2x-1}NO$ (x = 3–6)" *Moscow Univ. Chem. Bull.* **2013**, *68*, 171-174.

Patents

(1) **Eremin, D.**; Rodriguez, J. A.; Nelson, H. M.; Zhang, H. Fully Self-Driving Electron Diffraction Data Acquisition. U.S. Provisional Patent Application CIT-9295-P2, May 8, 2025.
(2) **Eremin, D.**; Nelson, H. M. A Software Suite for Autonomous Processing of Electron Diffraction Data. U.S. Provisional Patent Application CIT-9410-P, November 7, 2025.
(3) Mussetter, S. J.; Delgadillo, D. A.; **Eremin, D.**; Johnson, S.; Nelson, H. M. smPhase – Fragment-Based High-Throughput Molecular Replacement for Electron Diffraction Datasets of Complex Chemical Entities. U.S. Provisional Patent Application CIT-9412-P, November 12, 2025.
(4) Delgadillo, D. A.; Johnson, S.; **Eremin, D.**; Nelson, H. M. ArrayED: High-Throughput Crystallographic Characterization of Chemical Matter. U.S. Provisional Patent Application CIT-9413-P, November 13, 2025.

Non-peer-review publications

Eremin, D. B. "Highlights from Faraday Discussion: Reaction Mechanisms in Catalysis, UK, 17th–19th February 2021 (online)" *Chem. Commun.* **2021**, *57*, 6341-6345.

Invited Talks

(1) "Structure Without Barriers: Why Should Synthetic Chemists Care About MicroED?" *Organic Reactions and Processes Gordon Research Conference*, June 18, 2026, Stonehill College, Easton, MA, US
(2) "Structure and Reactivity in Complex Materials with Artificial Intelligence", January 21, 2025, Hosted by Prof. Michael Rose, *University of Texas at Austin*, Austin, TX, US
(3) "Teaching Machines Electron Diffraction" November 3, 2025, *MicroED Short Course*, New York Structural Biology Center, New York, NY, US
(4) "Teaching Machines MicroED: Fully Autonomous Structure Determination", August 15, 2025, Hosted by Prof. Christo Sevov, *Ohio State University*, Columbus, OH, US
(5) "Connecting the Dots in Scalable Droplet Synthesis", March 24, 2025, Reactivity in Microdroplets: Challenges and Potentials Symposium, *ACS National Spring Meeting*, San Diego, CA, US
(6) "A Journey to the Center of a Droplet", January 22, 2024, Hosted by Prof. Xin Yan, *Texas A&M University*, College Station, TX, US
(7) "A Journey to the Center of a Droplet", December 11, 2023, Hosted by Prof. Jahan Dawlaty, *University of Southern California*, Los Angeles, CA, US
(8) "A Tale of a Reaction: a Journey to the Center of a Droplet", November 21, 2023, Hosted by Prof. Olafs Daugulis, *University of Houston*, Houston, TX, US
(9) "Mass Spectrometry Journey to the Center of the Droplet", February 02, 2023, Hosted by Prof. Gregory Tochtrop, *Case Western Reserve University*, Cleveland, OH, US
(10) "Mass Spectrometry Journey to the Center of the Droplet", August 26, 2022, Hosted by Prof. Christian Malapit, *Northwestern University*, Evanston, IL, US
(11) "From the Solution to the Gas Phase: a Mass Spectrometry Guided Journey", July 22, 2022, Hosted by Prof. Catherine Costello, *Boston University*, Boston, MA, US

(12) "From the Solution to the Gas Phase: a Mass Spectrometry Guided Journey", June 10, 2022, Hosted by Prof. Gwendolyn Bailey, *University of Minnesota Twin Cities*, Minneapolis, MN, US

(13) "Mass Spectrometry in Mechanistic Studies and Detection of Elusive Intermediates", May 11, 2021, Hosted by Prof. Sachin Handa, *University of Louisville*, Louisville, KN, US

(14) "Mass Spectrometry in Mechanistic Studies and Detection of Elusive Intermediates", January 22, 2020, Hosted by Prof. Mark Gangelman and Prof. Ilan Marek, *Technion University*, Haifa, Israel.

(15) "Mass Spectrometry - a Key Tool for Investigation of Chemical Space", January 19, 2020, Hosted by Prof. Tomer Shlomi, *Technion University*, Haifa, Israel.

(16) "Mass Spectrometry as a Tool to Reveal Dynamic Catalytic Systems", May 3, 2019, *UK CatalysisHub Spring Conference*, Harwell Campus, Didcot, UK

(17) "Mechanistic Studies and Detection of Elusive Intermediates of Pd-Catalyzed Reactions by Mass Spectrometry" February 25, 2019, Hosted by Prof. Jana Roithová Laboratory, *Radboud University*, Nijmegen, The Netherlands

(18) "Development of Efficient Catalysts for Organic Transformations Based on Transition Metal Complexes and Nanoparticles", April 4, 2017, Hosted by Prof. Jin-Chong Tan, *University of Oxford*, Oxford, UK

Key Presentations

- Eremin D.B., Nelson H.M., "Teaching Machines MicroED: Fully Autonomous Structure Determination", *ACS National Fall Meeting*, August 17 - 21, 2025, **Washington, D.C., US** (*Oral*)
- Eremin D.B., Fokin V.V., "How Does Copper Change the pKa of Alkynes in Water?", *ACS National Fall Meeting*, August 13 - 17, 2023, **San Francisco, CA, US** (*Oral*)
- Eremin D.B., Aggarwal S., Vargas K., Hiller S., Grotsch K., Sundstrom S., Birch J., Nelson H., Fokin V.V., "Discovery of [3+2] Cycloaddition Reactions Enabled by Mass Spectrometry", *Organic Reactions and Processes Gordon Research Conference*, July 16 - 21, 2023, **Bryant University, Smithfield, RI, US** (*Poster*)
- Eremin D.B., Fokin V.V., "Selective C-H Oxidation of Heterocycles in Microdroplets and in Gas-Phase", *71th ASMS Conference on Mass Spectrometry and Allied Topics*, June 4 - 8, 2023, **Houston, TX, US** (*Poster*)
- Eremin D.B., Vargas K., Hiller S., Fokin V.V., "Water controls selectivity in triazole synthesis", *ACS National Fall Meeting*, August 21 - 25, 2022, **Chicago, IL, US** (*Oral*)
- Eremin D.B., Fokin V.V., " On-Water Control of Chemoselectivity in 1,4-Disubstituted Triazole Synthesis: Br vs SO_2F ", *Organic Reactions and Processes Gordon Research Conference*, July 17 - 22, 2022, **Bryant University, Smithfield, RI, US** (*Poster*)
- Eremin D.B., Vargas K.M., Hiller S.R., Fokin V.V., "From the Bulk to the Interface: Controlling Selectivity of Cycloadditions", *38th Reaction Mechanisms Conference*, June 12 - 15, 2022, **Boulder, CO, US** (*Poster*)
- Eremin D.B., Fokin V.V., " Dual Electrospray Enables Chemoselectivity Switch in Triazole Formation On-Water", *69th ASMS Conference on Mass Spectrometry and Allied Topics*, October 30 - November 4, 2021, **Philadelphia, PA, US** (*Poster*)
- Eremin D.B., Ananikov V.P., "Mass Spectrometry in Understanding Dynamic Phenomena in Catalytic Processes: a Case of the Mizoroki-Heck Reaction", *Reaction mechanisms in catalysis: Faraday Discussion*, February 17 - 19, 2021, **online, United Kingdom** (Poster)
- Eremin D.B., Ananikov V.P., "Mass Spectrometry as a Tool to Reveal Transformations in the Cross-Coupling Reactions", *Organometallic Chemistry Gordon Research Seminar & Gordon Research Conference*, July 6 - 12, 2018, **Salve Regina University, Newport, RI, US** (*Poster*)
- Eremin D.B., Ananikov V.P., "Dynamic Catalysis with Labile Pd/NHC Framework", *Organic Reactions and Processes Gordon Research Conference*, July 15 - 20, 2018, **Stonehill College, Easton, MA, US** (*Poster*)
- Eremin D.B., Ananikov V.P., "Mechanistic Studies and Detection of Elusive Intermediates of Pd- Catalyzed Reactions by Mass Spectrometry", *21st International Symposium on Homogeneous Catalysis (ISHC-2018)*, July 8 - 13, 2018, **Amsterdam, The Netherlands** (*Oral*)
- Eremin D.B., Ananikov V.P., "Direct ESI-MS study of catalysts activation and the Heck reaction mechanism", *Young scientist conference, Winter School in Organic Chemistry (WSOC-2018),* January 19 - 23, 2018, **Krasnovidovo, Moscow region** (*Oral*)

- Eremin D.B., Ananikov V.P., "A New Approach for Revealing Superatomic Cores in Transition Metal Complexes by ESI-MS/MS", *21st International Mass Spectrometry Conference (IMSC-2016),* August 20 - 26, 2016, **Toronto, ON, Canada** (*Oral*)
- Eremin D.B., Ananikov V.P., "ESI-MS and ESI-MS/MS study of transition metals acetylacetonates solutions", *VI National Conference "Mass Spectrometry and Applied problems" (The VII RMSS meeting)*, October 13 - 17, 2015, **Moscow** (*Oral*)

Service to Professional Community

- Peer-review activity: *Sci. Adv., PNAS, ACS Cent. Sci., Angew. Chem., J. Org. Chem., Chem. Eur. J., ChemCatChem, PCCP, Small Structures, New J. Chem., Chem. Select, Eur. J. Inorg. Chem, Appl. Organomet. Chem., Rapid Commun. Mass Spectrom*. (>150 peer reviews)
- *Review Panel Member* at CHE Research Instrumentation Program, NSF (2025)
- *Vice-Chair* for Los Angeles Metropolitan Mass Spectrometry (LAMMS) Interest group (2026)
- *Secretary/Treasurer* for MicroED Scientific Interest Group of ACA (2026 – 2027)
- *Program Committee Member* for ASMS National Meeting: 2023 (Houston); 2024 (Anaheim), 2025 (Baltimore)
- *Chair* for Los Angeles Metropolitan Mass Spectrometry (LAMMS) Interest group (2025)
- *Early Career Advisory Board Member* for *Chemistry – A European Journal* (2025 – present)
- *Treasurer* for Los Angeles Metropolitan Mass Spectrometry (LAMMS) Interest group (2022 – 2024)
- *Early Career Advisory Board Member* for *ChemistrySelect* (2022 – 2026)
- *Early Career Advisory Board Member* for *Mendeleev Communications* (2020 – present)
- *Discussion Leader* at Organic Reactions and Processes Gordon Research Conference, July 20-25, 2025, Bryant University, Smithfield, RI, US
- *Discussion Leader* at 73rd ASMS Conference on Mass Spectrometry and Allied Topics, June 1-5, 2025, Baltimore, MD, US
- *Discussion Leader* at ACS National Fall Meeting, August 13-17, 2023, San Francisco, CA, US
- *Discussion Leader* at 38th Reaction Mechanisms Conference (RMC2022), June 12-15, 2022, Boulder, CO, US
- *Discussion Leader* at International Conference "Catalysis and Organic Synthesis" (ICCOS-2019), September 15-20, 2019, Moscow
- *Assistant* at International Conference "Molecular Complexity in Modern Chemistry" (MCMC-2014), September 13-19, 2014, Moscow
- *Assistant* at XII International Conference on Nanostructured Materials (NANO-2014), July 13-18, 2014, Moscow

Internship experience

- "FT/ICR-MS training course on solariX 2XR", April 19-20, 2017, Bruker Daltonik GmbH., Bremen, Germany
- FELIX project #62003247 with EU support; FELIX 2, 6 shifts, February 25 – March 1, 2019, Radboud University, Nijmegen, The Netherlands

Funding

PI (*funded*): "*Single-Cell Mass Spectrometry Enabled by Orthogonal Charge-Tag/Fluorescent Dye Modification*", Agilent Postdoctoral Fellowship Grant, 04/2019-04/2021, $200,000 direct cost.

Mentoring and Teaching

Mentees

Graduate students (Fokin group)

Sydney R. Hiller – 2019–2023 Development of chemoselective synthesis of isoxazoles
Kevin M. Vargas – 2020–2024 On-water control of chemoselectivity in the triazole synthesis
Rudra Persaud – 2021–2024 New generation of covalent ligands for GPCR proteins

Graduate students (Ananikov group)

Liliya T. Sahharova – 2017–2018 Development of transformations with vinyl-sulfides
Ekaterina D. Sushchenko – 2017–2018 Activation of low-reactive substrates by dynamic M/NHC systems
Ekaterina A. Denisova – 2017–2018 Dynamics of the Pd/NHC systems

Undergraduate students (Nelson group)

Benjamin Rudo – SURF 2026 Caltech, Automated microED data indexing

Emma Fan – 2025–2026 Caltech, *Diffractomics*, data science tools for microED

Samuel Foxman – 2024–2026 Caltech, Web platform crowdsources electron diffraction data labeling

Hongyu Zhang – 2024– 2025 Caltech, Cloud-database for microcrystalline electron diffraction

Undergraduate students (Fokin group)

Diego Montano – 2020–2021 USC, Investigation of covalent protein complexes with mass spectrometry

Alexander Vu – 2020–2021 USC, DFT study of water-assisted transformations, on-water synthesis of isoxazoles

Undergraduate students (Ananikov group)

Daniil A. Boiko – 2016–2019 Lomonosov Moscow State University student; Course projects in Inorganic chemistry and Analytical chemistry; 2017 project selected as "The best Course projects in Analytical Chemistry" by Analytical Chemistry Division, Chemistry Department

Nikita S. Gushchin – 2016 Lomonosov Moscow State University student; Introduction to the lab work

Nikita S. Shlapakov – 2014–2015 Lomonosov Moscow State University student; Course projects in Analytical chemistry; project selected as "The best Course projects in Analytical Chemistry" by Analytical Chemistry Division, Chemistry Department

High school students (Nelson group)

Shayla Starr – 2025 – present Caltech, Gamification of mciroED

Developed courses

ChE 190: Special Problems in Chemical Engineering: Beyond Mass Measurement with Artificial Intelligence, 9 untis, Caltech, Division of Chemistry and Chemical Engineering

CHEM 599: Mass spectrometry in mechanism and structure investigation, 4 units, USC, Department of Chemistry

Teaching

ChE 190: Special Problems in Chemical Engineering: Beyond Mass Measurement with Artificial Intelligence, 3-0-6, Caltech, Division of Chemistry and Chemical Engineering, Spring 2026

CHEM 105A: General Chemistry, co-instructor with Prof. Thomas Bertolini, 4 units, USC, Department of Chemistry, Spring 2022

CHEM 565L: Advanced Practical NMR Spectroscopy, co-instructor with Prof. Travis Williams, 2 units, USC, Department of Chemistry, Spring 2022

CHEM 599: Mass spectrometry in mechanism and structure investigation, co-instructor with Prof. Valery Fokin, 4 units, USC, Department of Chemistry, Fall 2020, Fall 2021, Fall 2022

CHEM 599: Kinetics and Mechanism of Catalytic Reactions, co-instructor with Prof. Valery Fokin, 4 units, USC, Department of Chemistry, Fall 2023

Professional Membership

Member of American Crystallographic Association (ACA) (since 2025)

Member of American Society for Mass Spectrometry (ASMS) (since 2020)

Member of American Chemical Society (ACS) (since 2019)

Associate Member of Royal Society of Chemistry (RSC) (since 2017)